

LUCA MITTINO CASTRO

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EDUCATION

Northeastern University — B.S. Combined Mechanical Engineering & Design

Expected May 2028

Boston, MA · Dean's List (2025) Activities: ASME, NURobotics VexU Mechanical Design

Coursework: Statics, Physics I–II, Calculus I–III, Cornerstone of Engineering, Mechanics of Materials, Differential Eq and Linear Algebra, Thermodynamics |

American School of Madrid — International Baccalaureate Diploma

Graduated Jun 2024

Madrid, Spain · Graduated with Honors

EXPERIENCE

Mecwins | R&D Engineer Intern

Jun – July 2026 · Madrid, Spain

- Prototyped and designed a fully functional budget motorized microscopy platform using Adobe Inventor CAD from a 3D printer base by building a Python application to control an industrial camera and real-time motion hardware.
- Enabled hands-free, precise imaging by implementing an automatic autofocus algorithm and a pixel-to-micron calibration system for accurate quantitative measurement in Python.
- Used Inventors FEA simulations to ensure structural reliability, under stepper motor vibration, measured by a predicted 0.0375 mm maximum displacement and 0.116 MPa peak stress under the selected worst-case sample load

Tang Research Lab | Undergraduate Research Assistant

May – Jun 2026 · Boston, MA

- Analyzed droplet impact dynamics via high-energy X-ray imaging in the Tang Lab, characterizing splash behavior across substrates to inform predictive models, validated through particle-level trajectory tracking.
- Supported particle density separation and data collection on non-Newtonian shear-thickening fluids, isolating splash-suppression mechanisms measured across substrate comparisons.

Pursuit Aerospace | Manufacturing Quality Engineer CO-OP

Jan – June 2026 · Boston, MA

- Recognized by CEO and GM for enabling the highest single order outflow in company history by prioritizing inspections and coordinating quality operations across a critical production surge.
- Drove nonconformance rate from 27% to 6% a 78% improvement, by collaborating with engineering and manufacturing teams to identify and eliminate root-cause defect sources on an active production part.
- Sustained ~135 part-per-week inspection throughput with zero escapes by applying stylus, caliper, and optical comparator techniques against GD&T compliant engineering drawings.
- Prevented customer driven rework by serving as technical liaison between both engineering and quality teams, resolving manufacturability issues before production started during active customer audits.

Myocardial Solutions | AI Automation Intern

Jun – Aug 2025 · Madrid, Spain

- Cut patient record processing time down by 90% by developing an AI driven pipeline to automate database sorting and filing across multiple local hospital systems.
- Supported two government grant applications by authoring engineering justification sections covering system architecture and clinical workflow integration.

ENGINEERING PROJECTS

UAV Frame Design & Structural Analysis | Personal Project

Aug 2026 – Present

- Iterating three drone frame geometries (True-X, H-Frame, Hybrid-X) in SolidWorks using ANSYS FEA and CFD to optimize stiffness-to-weight ratio against a defined payload and mission profile.
- Validating simulation results by fabricating and load-testing 3D-printed prototypes.

Fuel-Optimal Powered Descent Guidance Algorithm | Personal Project

Jun – Sept 2026

- Built a rocket landing guidance algorithm in Python by applying convex optimization to reformulate a non-convex thrust constraint into a solvable fuel-minimization problem.
- Validating trajectory outputs by modeling vehicle dynamics and descent profiles in MATLAB.

Catch the Sun | Northeastern Engineering Cornerstone Project

Feb – May 2025

- Led mechanical design for a 4-person team, delivering a climate-education interactive exhibit on schedule using SolidWorks, AutoCAD, laser cutter, 3D printing, and soldering within strict size and transport constraints.
- Achieved zero failures across 3 live demonstrations by integrating a MATLAB-based UI with an C++ control system for real-time exhibit interactivity.

LEADERSHIP

Scouts of America | Senior Patrol Leader/Patrol Leader (Eagle Scout)

Sep 2019 – Jun 2024

- Directed weekly operations for a 55-person troop by leading meetings, coordinating community service projects, and mentoring junior patrol leaders through structured development cycles.

SKILLS

CAD / Sim: SolidWorks (CSWA), ANSYS FEA & CFD, Siemens NX, AutoCAD, Adobe Inventor | **Programming:** C++, Arduino, MATLAB, Python

Manufacturing: GD&T, 3D Printing, Laser Cutting, Soldering, Iterative Prototyping | **Languages:** English & Spanish (native)

Hobbies: Music, Soccer, Weightlifting, Rockets, Hiking, Traveling, Eating